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Methane and hydrogen production from cotton waste by dark fermentation under anaerobic and micro-aerobic conditions

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ABSTRACT

The dark fermentation (DF) of organic textile (e.g. cotton) wastes is discussed in this paper. The influence of micro-aeration on hydrogen and methane production is studied. Fermentation was carried out under the following conditions: load 5 g VSS/L (Volatile Suspended Solids), the pH was varied in the range from 6.23 to 7.8; oxygen was added in small quantities. The oxygen flow rates (OFRs) were varied between 1 and 4.6 mL/h. The highest yields of methane (0.78 L/g VSS) were obtained under anaerobic conditions and neutral pH (7.54), while the production of hydrogen (1.1 mL/g VSS) peaked under an oxygen flow rate of 4.6 mL/h and a pH value of 6.4. Hydrogen and methane yield and the peak days of their production from various substrates (such as cotton waste, sour cabbage, wheat straw, and potato wastes) were compared and discussed. Cotton waste is a potential source of methane, but is substantially less efficient than potato waste.

1. Introduction

Dark fermentation (DF) is a type of anaerobic digestion process, which converts various substrates into hydrogen, carbon dioxide and low organic acids [1]. Before viable DF may be achieved on an industrial scale, methane generation, the process that removes hydrogen from the biogas, must be inhibited, usually by a proper pretreatment of the substrate and/or inoculum. Although, Lakaniemi et al. [9] and Li et al. [10] have shown that, for some substrates, dark fermentation occurs without pretreatment.

Hydrogen is a prime raw material for many industrial processes [2], and a possible 'green' replacement of conventional sources of energy [3]. Although the storage of hydrogen remains a problem [7] and this factor has inhibited the development of biohydrogen technologies [8], demand for hydrogen is growing. Thus an increased variety of potential pathways for its production is desirable, e.g. pyrolysis [4], dark fermentation [5] or microbial electrolysis [6].

There have been many attempts to industrialize the production of hydrogen by DF using batch- and continuous reactor-types but so far none have been able to scale-up the process of dark fermentation; the processes are too unstable and hydrogen production falters after some time [11]. Another issue is that the hydrogen conversion rate is too low [12].

Although biomethane production via anaerobic digestion is a much better known process, it also needs some improvements [13]. The process usually proceeds in the mesophilic range (33°C–40 °C), but recent achievements in the efficient production of more valuable byproducts such as lactic acid were obtained under thermophilic conditions (55°C–60 °C) [14]. The optimal pH for hydrogen production by dark fermentation varies depending on the substrate is in the range 5.0–6.0, while for methane production it is within the range of 7.0–9.0 [15].

As not many investigations have been done for hydrogen and methane production from cotton waste, a principal objective of this research is to search for the optimal value of pH and microaeration rate [16].

Aeration (or microaeration) is the technique of adding air, applied in many biological processes such as composting, photofermentation, microbial electrolysis and even anaerobic digestion [17]. The addition of 2–8% oxygen facilitates the production of hydrogen in the photofermentation process and has also been referred to as microaerobic dark fermentation in Refs. [18]. According to Nguyen et al. [19] and Samson et al. [20], microaeration should enhance methane production and increases the yield of microbial electrolysis. Besides, the addition of oxygen in small amounts is used in anaerobic digestion to inhibit hydrogen sulphide formation in batch [21], semibatch [22] and continuous reactors [23]. According to Krayzelova et al. [24], microaeration/microoxygenation occurs when the addition of air/oxygen to anaerobic fermentation is less than 1 L O_2 per 1 L of feed. The optimal

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microaeration rate depends upon the substrate and reactor type [25]. Microaeration also improves the rate of hydrolysis and increases the degree of hydrolysate conversion [26]. Thus, it would appear to be worth investigating which process, methanogenesis or hydrogenesis (both being anaerobic), is more sensitive to the presence of oxygen, which may facilitate the control of these processes during dark fermentation.

It is also worth comparing changes in hydrogen production during dark fermentation of various substrates to discern which group of substrates is more efficient; self-acidizing or neutral. An investigation of the influence of a low pH substrate showed that neither the pKa nor the pH of the substrate is a decisive parameter without the determination of the VSS of the substrate, see e.g. Refs. [27–30].

One of the problems associated with lignocellulose (such as wheat straw) pretreatment relates to the presence of lignin [29] and its derivatives [30]. Among the lignin derivatives that inhibit the fermentation process are hydroxymethylfurfural (HMF) [31] or vanillin [32]. Therefore an appropriate substrate for dark fermentation should contain mainly simple carbohydrates and should not contain lignin [33]. Lack of lignin should be a property of the substrate alone [5] or should be obtained through appropriate pretreatment [34]. Therefore, cotton waste, due to its high cellulose and low lignin content [35], may be considered as a promising material for fermentation [36,37]. Besides, it may lead to potentially higher hydrogen yield than wheat straw or corn straw [38]. However, it was reported [39], that under some hydrothermal or acidic cotton pretreatments a high concentration of HMF was found.

Hydrogen production from other lignocellulosic materials like wheat straw was studied by Nasirian et al. [40,41]. They obtained 94% of the theoretical hydrogen yield as calculated in Ref. [38] after pretreatment (using 2% sulphuric acid) and at a very low pH value of 2.4, i.e. under a lower pH value than it is usually required for dark fermentation. Sulphuric acid is proposed as the least corrosive and most efficient substance available for the enhancement of lignocellulose fermentation by anaerobic digestion [42] or dark fermentation [43].

In this paper, methane and hydrogen production (preceded by substrate pretreatment) from cotton waste during dark fermentation under microaerobic conditions is analysed. To the best of our knowledge, there are no reports available to date concerning the batch dark-fermentation process using cotton waste (clothes) as a substrate and on its dependence on the microaeration rate. There are some studies concerning methane production under microaerobic conditions, but none of them focus on hydrogen production [44]. The comparison with sour cabbage was made to check whether a self-acidizing material is more valuable for dark fermentation than pH-neutral cotton waste. A comparison of the results of cotton and wheat straw fermentation with those of sour cabbage [45,46] and other substrates [10] such as asbestos [47] or depackaged food waste [48] is presented in the article.

2. Material and methods

The fermentation process of cotton waste was performed in 2 dm^3 glass reactors with a working volume of 1.2 dm^3 - see Fig. 1. The inoculum used for the experiments originated from a mesophilic digester mainly used to treat maize silage and manure (Table 1). The tested substrates (5 g VSS/L) of cotton waste were placed in the reactors with the inoculum. In order to maintain the appropriate conditions, they were kept in a water bath under mesophilic conditions (38 ± 2 °C). Before fermentation, the reactors were flushed with nitrogen to maintain strictly anaerobic conditions at the beginning of the process. The gas produced by every fermenter was collected in a cylindrical vessel filled with water and a barrier liquid.

The substrate, cotton waste and wheat straw, was finely shredded and hydrolysed using a 0.1 M acidic (HCl) solution for 2 h before being placed into the reactor. Hydrolysis was performed at room temperature in a closed 500 mL vessel using 250 mL of 0.1 M HCl for every 60 g of cotton waste. Next, the cotton waste was removed from the vessel and neutralized with deionized water to a pH value of 7.2.

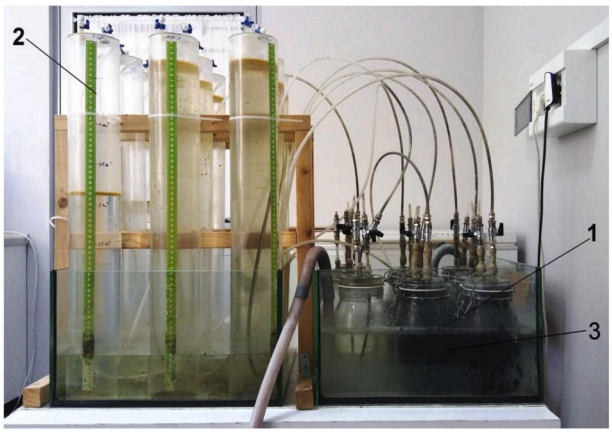


Fig. 1. Fermentation setup used in the experiment: 1. Glass reactors, 2. Cylindrical vessel for collecting biogas, 3. Water bath chamber under mesophilic conditions (38 ± 2 °C).

Table 1
Physicochemical characteristics of the inoculum and substrates used in various tests.

Material	pH	TS [%FM]	VSS [%TS]
Inoculum	8.2	1.09 ± 0.03	45.35 ± 1.03
Cotton	–	100	79 ± 1.06
Sour cabbage	4.6	6.99 ± 0.02	89.3 ± 1.2
Wheat straw	8.2	8.02 ± 0.02	65 ± 1.11
Potato waste boiled	5.83	9.05 ± 0.02	35 ± 1.11
Potato waste unboiled	5.7	7.05 ± 0.02	33 ± 1.14

To avoid inhibiting the dark fermentation reported by Sasaki et al. [39] no other pretreatment was performed. After adding the substrates to the inoculum, the pH value of the fermenting mixture was lowered from 7.8 to 6.2 with the addition of 50% sulphuric acid. Sour cabbage was only milled. Potato waste from restaurants: both boiled and unboiled were milled and used as a substrate for comparison. Before fermentation, the pH, total solids (TS) and volatile suspended solids (VSS, [%TS]) were determined for the fresh mass [FM] of the substrates.

In the experiments, the following oxygen flow rates (OFRs) were applied: 0.55, 1, 1.5, 2.1 and 2.6 ± 0.01 mL/h for neutral pH 7.8 (cotton waste). In the case of an acidic low pH value, OFR values of approximately 1, 2.7 and 4.6 ± 0.01 mL/h were applied to observe if microaeration improves hydrogen production. The OFR was increased also above the level of micro-oxygenation (more than 8%) to discern the upper level of oxygen concentration where hydrogen production may still be observed. The oxygen was added twice a day until the fermentation process was stopped, after 24–26 days. The oxygen was added for a duration of approximately 2 s. The pH of the fermenting mixture was kept stable using NaOH 0.1 M solution and 50% H_2SO_4, and pH was monitored once a day. The volume of biogas produced was measured every day. The studies for the load of 5 g VSS/L of sour cabbage were continued and extended using the conditions described in Ref. [45], for OFR 3.3 and 4.1 mL/h under pH of 7.54. Later, the results of cotton wastes fermentation were compared with those of sour cabbage obtained in this and previous studies. In the case of wheat straw, the OFR ranged from 0 to 4 mL/h under neutral conditions and from 0 to 4.5 mL/h under acidic conditions; the pH value was varied from 8.3 to 6.0. For potato wastes OFR was 1 mL/h and pH 6.3.

The qualitative and quantitative assessment of the gases was performed and determined in two stages. During the first, the gas was assessed using a portable biogas analyzer (GA5000, Geotech), with the volume of biogas in the cylinder being at least 0.45 dm^3.

The analyser poses ATEX II 2G Ex ib IIA T1 Gb (Ta = −10 ÷ +50 °C),

IECEx and CSA quality certifications and UKAS ISO 17025 calibration certificate. The equipment allows the measurement of CH_4, CO_2, O_2, H_2 and H_2S in the ranges 0 ÷ 100%, 0 ÷ 100%, 0 ÷ 25%, 0 ÷ 1000 ppm and 0 ÷ 5000 ppm, respectively. Calibration of the device was performed twice a week. During the second stage, when hydrogen concentration was above 1000 ppm, the gas was assessed using a gas chromatograph (GC) with a thermal conductivity detector and argon as a carrier (gas flow rate was 0.6 mL/h). A Silco packed column Restek® with characteristics of 2 m/2 mm ID 1/8″ OD Silica was used. All of the experiments were carried out in triplicate, the mean values for biogas measurements are reported.

The hydrogen (methane) production time was defined as the period from the start of experiment until the day when the hydrogen (methane) production declines below 1% of the value of the total hydrogen (methane) production achieved earlier in the process.

3. Results and discussion

GC analysis allowed for the determination of the methane, hydrogen, carbon dioxide, oxygen and nitrogen concentrations. Biogas production was measured for 25 or 26 days, but in the case of microaerobic fermentation (with a pH of 7.54 for sour cabbage, 8.3 for wheat straw and 7.54 for cotton) fermentation was continued until the 27th day. Figs. 2–4 present the time evolution of methane produced during DF of cotton waste, sour cabbage and wheat straw (with a load of 5 g VSS/L under various conditions).

In the case of cotton waste, the process was performed under different pH values: 7.54 (neutral) or acidic pH 6.2 or 6.4 (after H_2SO_4 addition) and oxygen flow rates in order to determine the influence of these conditions on the production of methane and hydrogen. Methane production under anaerobic conditions (pH 7.54) was much higher

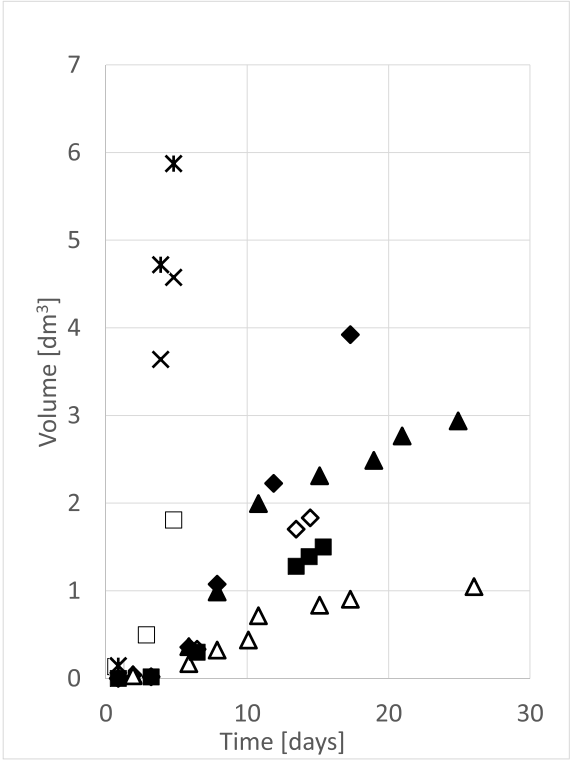


Fig. 2. Cumulative methane production from cotton waste and sour cabbage for loads of 5 g VSS/L: ◆ cotton wastes anaerobic pH 7.54; ▲ cotton wastes OFR 1 mL/h, pH 6.23; Δ cotton wastes OFR 2.1 mL/h, pH 7.54; × potato boiled wastes OFR 1 mL/h, pH 6.23; * asterisks potato unboiled wastes OFR 1 mL/h, pH 6.23; ■ dry wheat straw OFR 27 mL/h, pH 8.3; ◊ dry wheat straw OFR 2.8 mL/h, pH 8.3; □ sour cabbage anaerobic pH 7.54 [31].

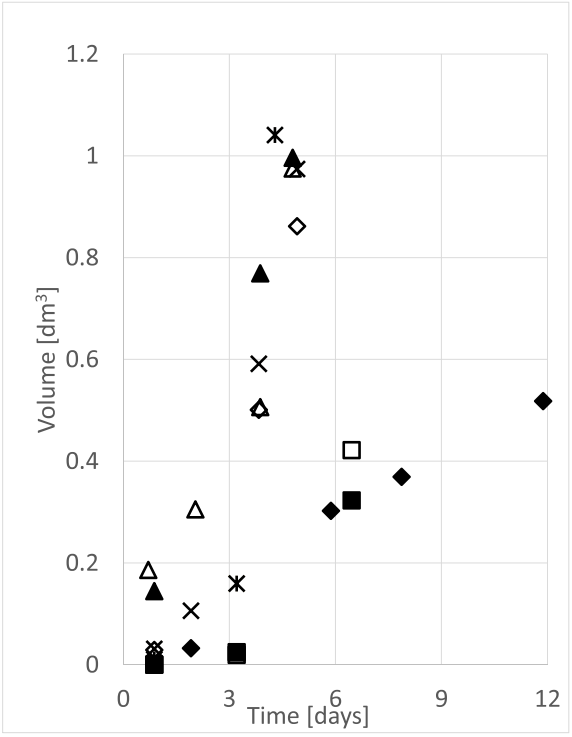


Fig. 3. Cumulative methane production from, potato wastes, wheat straw for loads of 5 g VSS/L: ■ dry wheat straw OFR 5.1 mL/h, pH 6.0; ▲ hydrolysed wheat straw anaerobic pH 7; □ dry wheat straw OFR 3.5 mL/h, pH 8.3; ◊ hydrolysed wheat straw OFR 1.9 mL/h pH 6.21; × dry wheat straw pH 6.23 OFR, 1.3 mL/h; Δ sour cabbage OFR 0.63 mL/h pH, 7.54 [31]; ◆ cotton wastes OFR 0.55 mL/h pH 7.54; * dry wheat straw OFR 3.4 mL/h pH 6.23.

(3.92 L after 17 days of fermentation). The ratio of volume concentrations of the main biogas components CH_4:CO_2 was 73.8:26. At the same pH value of 7.54 but under OFR 1.5 mL/h, the methane yield was 1.74 L (after ~ 27 days). In the case of the lower pH value, after 25 days the methane production was 0.94 and 2.94 L, for OFR = 4.6 mL/h (pH 6.4) and OFR = 1 mL/h (pH 6.23), respectively - see Fig. 2. Moreover, the ratio of volume concentrations of the main biogas components CH_4:CO_2 slowly decreased from 63.5:36.5 to 59.6:40.4 as the OFR was increased from 1 mL/h to 4.6 mL/h, respectively; the effect was only slightly more evident than in the case of Aslanzadeh et al. (55–60%) [44,49,50].

The results obtained for cotton were compared with earlier studies for the fermentation of sour cabbage at a pH value of 7.54 [45] and wheat straw. In the cases of hydrolysed wheat straw and milled straw, there was no biogas production under strictly anaerobic conditions (both under neutral and acidic pH conditions). When comparing with anaerobic cotton digestion (where biogas production was much more significant) one can conclude that the lignin content in wheat straw is high enough to block methane production without microaeration. Under microaerobic conditions, both neutral and acidic pH conditions, fermentation lasted for 4 days (acidic), and 7 days (neutral); a longer duration of methane production (about one week) may be observed at pH 8.3 than was the case at pH 6.23 see Table 2.

In the case of anaerobic fermentation of sour cabbage, the methane production was significantly lower than for cotton, at least at the initial stage of the process (up to 5 days). Neither result confirms the conclusions of Nguyen et al. [19] concerning methane production enhancement under microaeration conditions. Their results indicated that microaeration improved methane production while in this study only hydrogen production increases at some OFRs. Lowering the pH at a higher OFR value reduces methane production in comparison to anaerobic neutral conditions, which is in agreement with the previous results of methane production from lignocellulosic wastes [51].

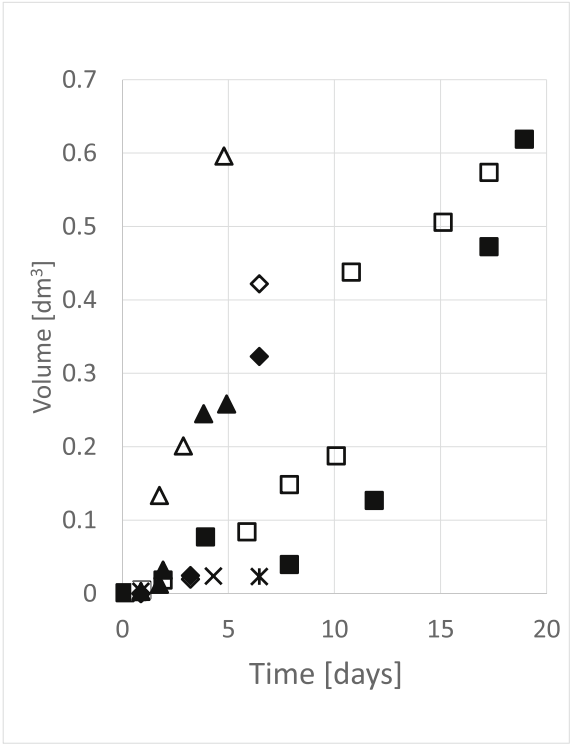


Fig. 4. Cumulative methane production from wheat straw and sour cabbage for loads of 5 g VSS/L; × dry wheat straw OFR 1.5 mL/h, pH 6.3; ▲ hydrolysed wheat straw OFR 5.2 mL/h, pH 6.0; ⋆ dry wheat straw OFR 1.4 mL/h, pH 6.3; ◊ dry wheat straw OFR 3.5 mL/h pH 8.3; Δ sour cabbage OFR 3.3 mL/h, pH 7.54; ◆ dry wheat straw OFR 2.1 mL/h, pH 8.3; ■ cotton wastes OFR 4.6, pH 6.4; □ cotton wastes OFR 2.6 mL/h, pH 7.54.

Assessing the optimal pH for dark fermentation of different stock is important; see reports on neutral pH for duckweed [52], potatoes [53] and basic pH for cotton straw [10]. However, methane production under microaeration conditions is higher in acidic rather than in neutral conditions. In the case of anaerobic fermentation of sour cabbage, the methane concentration in biogas was significantly higher than for cotton, at 73.8% ± 2%.

During fermentation of boiled potato under pH of 6.3 and OFR of 1 mL/h, 4.58 L of methane was produced, while from unboiled potato at pH of 6.03 and an OFR of 1 mL/h, 5.87 L of methane was produced.

Hydrogen production during cotton waste fermentation is shown in Figs. 5–6. The production rate was rather low: under anaerobic conditions and pH 7.8 (reached after 415 h - more than 17 days) approximately 0.37 mL; and it increased (after 25 days) when microaeration was included up to 2 mL of H_2 (6-fold increase) for low oxygen flow rates (0.55 mL/h). Under OFR equal to 1.5 mL/h production reached 1.35 mL of hydrogen after 25 days. The decrease of pH under optimal OFR (4.6 mL/h) facilitated much higher yield of 5.5 mL of hydrogen, while under an OFR of 1 mL/h it fell to 2 mL.

The hydrogen production time is almost as long as the methane production time. In the case of a neutral pH and OFR = 1.5 mL/h, the hydrogen production process was one day shorter than it was for methane. However, both gases were present in biogas during all of these days.

Hydrogen production from cotton waste during DF, is in agreement with the other investigated cases [54], it increases with decrease of pH value up to the optimal range of 5.0–6.0 [55]. The shift of the optimal pH into acidic conditions agrees also with the results [56,57]. Hydrogen production from cotton waste for an OFR of 4.6 mL/h and a pH value of 6.4 is much higher than under anaerobic conditions (11 times) and microaerobic conditions at neutral pH values (up to 3 times). However, hydrogen production under neutral conditions at an OFR of 1.5 mL/h is slightly higher than for acidic conditions and at OFR ranging from 1 mL/h to 2.7 mL/h.

A shift in hydrogen production with a lower pH value may also occur due to a lowering of the stability of cellulose, thereby improving digestion by the bacteria [58]. Under microaeration conditions hydrogen production can last 9 days longer than in the case of anaerobic conditions.

The data are compared with the results for the fermentation of sour cabbage under the same load of 5 g VSS/L [59] - see Figs. 5–6 and Table 2. Under similar OFR and pH values, cotton wastes are a slightly better hydrogen source. Therefore, it may be deduced that non-lignin substrate is a more valuable hydrogen production source. However, the much shorter time of hydrogen production (~6 days) for sour cabbage than for cotton waste (up to 27 days depending on aeration conditions) should be noted. For sour cabbage, both hydrogen and methane production ceases after approximately 6 days.

Hydrogen and methane production under mesophilic conditions from different substrates are compared in Table 2. The results relating to thermophilic conditions are omitted here as they lead to very different yields [14]. In the case of our own results, we also present the hydrogen and methane peak days. On peak days, the production of methane or hydrogen (per gram of VSS) is much higher than on other days. In the case of cotton waste, the peak days for hydrogen and methane production are different. The peak day of hydrogen for cotton waste under anaerobic conditions occurred on the 12th day, while under microaeration conditions, it occurred on the 15th day. Methane fermentation peaked for cotton waste on the 17th day (anaerobic) and on the 18th day (microaeration). It may be discerned that, for DK of sour cabbage (under microaeration conditions), the peak day of hydrogen occurred earlier than for methane. In the case of cotton, the peak day of hydrogen occurred after the passing of 0.66 (anaerobic) and 0.60 (microaeration) of the overall fermentation time.

For sour cabbage, the peak day of methane production occurred on the 5th day; this was independent of the concentration and microaeration conditions. The peak day of hydrogen production for 5 g VSS/L occurred on the 1st day of fermentation. In the case of sour cabbage, at 10 g VSS/L microaeration shifted the peak day of hydrogen to the 6th day of fermentation (two days earlier than under anaerobic conditions).

The results of the fermentation of various substrates shows that cotton waste is an inferior source of methane compared to sour cabbage and the organic portion of municipal wastes [60,61]. It is generally assumed that during dark fermentation, methane is not produced, although this assertion was rejected by the authors of ref. [9], they also claim that for hydrogen production, it is not necessary to stress the bacteria in the inoculum.

Hydrogen production from milled wheat straw was very low compared to that of cotton. In the case of a neutral pH of 8.3, it ranged from 0.8 to 3.5 mL/h. At an OFR of 4 mL/h, hydrogen production ceased. As can be seen in Fig. 6, hydrogen production from wheat straw has a longer duration for an OFR of 2.7 mL/h than for 1.6 mL/h.

During fermentation of boiled potato under pH value of 6.3 and an OFR of 1 mL/h, 167 mL of hydrogen was produced, while for unboiled potato waste at a pH of 6.03 and an OFR of 1 mL/h, only 48 mL of hydrogen was produced.

The peak day data for hydrogen production (Table 2), shows the volume of hydrogen yield for the day of the highest production. Hydrogen yield is the volume of hydrogen produced per gram of substrate. In some cases, as with wheat straw at a pH of 6.3 and an OFR of 3.8 mL/h, a high rate of hydrogen production was only achieved during the peak day while during other days only small fraction of hydrogen was produced. Therefore, the hydrogen peak day yield is higher than the daily average one. It may be discerned from Fig. 6 and Table 2, that there are days with sudden surges of hydrogen production, which are much higher than the usual growth in hydrogen volume. In the case of untreated inoculum, it may be observed in Table 2 that the hydrogen peak

Table 2
Methane and hydrogen production yield for different substrates.

Substrate	Substrate load g VSS/L	Inoculum	Conditions	CH_4 Yield mL/g VSS	CH_4 Peak day mL/day	H_2 Yield mL/g VSS	H_2 Peak Day mL/day	Refs
Sour cabbage	10	Raw agricultural digestate	Anaerobic T = 38 °C pH 7.8	550	197 5th day	6.58	0.83 8th day	[46]
Sour cabbage	10	Raw agricultural digestate	Microaerobic T = 38 °C pH 7.8 2.5 mL/h	233.3	79 5th day	40.75	1.33 6th day	[46]
Sour cabbage	5	Raw agricultural digestate	Anaerobic T = 38 °C pH 7.8	141.7	70 3rd day	0.36	0.05 1st day	[45]
Sour cabbage	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 7.8 0.62 mL/h	162.5	90 5th day	1.08	0.1 1st day	[45]
Cotton waste	5	Raw agricultural digestate	Anaerobic T = 38 °C pH 7.8	653.3	93.33 17th day	0.063	0.013 12th day	This study
Cotton waste	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 7.8 OFR 0.55 mL/h	290	56.17 6th day	0.317	0.275 6th day	This study
Cotton waste	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 7.8 OFR 1.5 mL/h	143.3	45 6th day	0.044167	0.0453 11th day	This study
Cotton waste	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 7.8 OFR 2.1 mL/h	112.5	46.33 11th day	0.0225	0.00133 11th day	This study
Cotton waste	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 7.8 OFR 2.6 mL/h	95.8	41.66 11th day	0	–	This study
Cotton waste	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 6.4 OFR 4.6 mL/h	158.3	650 18th day	0.917	0.33 15th day	This study
Cotton waste	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 6.3 OFR 2.7 mL/h	98.33	185	0.217	0.058 2nd day	This study
Cotton waste	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 6.23 OFR 1 mL/h	483.3	1000 11th day	0.317	0.125 11th day	This study
Boiled potato waste	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 6.23 OFR 1 mL/h	231.7	572 3rd day	27.5	20.5 3rd day	This study
Unboiled potato waste	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 6.23 OFR 1 mL/h	145.8	572 3rd day	0.8	1.67 4th day	This study
Hydrolysed wheat straw	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 6.23 OFR 4.6 mL/h	146.7	418 4th day	31.67	22.67 3rd day	This study
Hydrolysed wheat straw	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 6.23 OFR 3.1 mL/h	42.5	213 4th day	18.33	6.27 3rd day	This study
Hydrolysed wheat straw	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 7 OFR 3.1 mL/h	165.8	104 4th day 624 4th day 2.8 28 4th day This study	2.33 624 4th day 2.8 28 4th day This study	1.15 1st day 624 4th day 2.8 28 4th day This study	This study 624 4th day 2.8 28 4th day This study
Milled wheat straw	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 6 OFR 3.4 mL/h	166.7	146.67 3rd day	0.002917	0.001167	This study
Milled wheat straw	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 6.3 OFR 3.8 mL/h	3.916	0.00333	0.000917	0.000833 4th day	This study
Milled wheat straw	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 6.3 OFR 1.4 mL/h	173.3	146.883 4th day	0.001	0.00043 4th day	This study
Milled wheat straw	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 8.3 OFR 0.8 mL/h	241.7	163.33 3rd day	0.0013	0.00042 4th day	This study
Milled wheat straw	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 8.3 OFR 1.6 mL/h	214.2	135 4th day	4.67	6.3 14th day	This study
Milled wheat straw	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 8.3 OFR 2 mL/h	159.2	137.13 4th day	0.011	0.029 2nd day	This day

(*continued on next page*)

Table 2 (*continued*)

Substrate	Substrate load g VSS/L	Inoculum	Conditions	CH_4 Yield mL/g VSS	CH_4 Peak day mL/day	H_2 Yield mL/g VSS	H_2 Peak Day mL/day	Refs
Milled wheat straw	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 8.3 OFR 2.1 mL/h	53.3	49.67 6th day	0.0033	0.0001 7th day	This study
Milled wheat straw	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 8.3 OFR 2.7 mL/h	305	228 3rd day	0.0025	0.0006 4th day	This study
Milled wheat straw	5	Raw agricultural digestate	Microaerobic T = 38 °C pH 8.3 OFR 3.5 mL/h	70	67–3rd day	0.002925	0.0017 3rd day	This study
Diary manure	5	Raw agricultural digestate	Anaerobic T = 35 °C pH 6.9	350	282	–	–	[62]
Organic fraction of municipal solid waste	5	Raw agricultural digestate	Anaerobic T = 37 °C pH 6.9	690	–	–	–	[63]
Food wastes	8	Raw agricultural digestate	Anaerobic T = 35 °C pH 7.3	540	400 1st day	–	–	[64]
Switchgrass	10	Raw Anaerobic digestate	Anaerobic T = 35 °C pH 7.3	257	400 1st day	–	–	[65]
Elephant grass hydrolysate	12.5	Raw agricultural digestate	Anaerobic T = 35 °C pH 7.0	96	–	–	–	[66]
Cotton stalk hydrolysate	40 (sugar)	Bacteria from wild carp intestine	Anaerobic T = 37 °C pH 8.0	–	–	179	750	[10]
Reed canary grass hydrolysate	5	Digestate (municipal activated sludge)	Anaerobic T = 38 °C pH 4.6	–	–	36	–	[9]
Hydrothermally asbestos to glucose of proportion 1:6	–	Thermal treated digestate from biogas plant	Anaerobic T = 35 °C pH 4.5	–	–	0.013	–	[47]
Pretreated grass	5	Cultures enriched from cracked cereal	Anaerobic T = 35 °C pH 7.0	–	–	4.39	–	[67]
Raw cassava starch	10 starch	Anaerobic activated sludge	Anaerobic T = 35 °C pH 7.0	–	–	240	–	[68]
Pretreated wheat straw	15	Microflora from cow dung compost	Anaerobic T = 36 °C pH 6.5	–	–	22.9	–	[69]
Sunflower stalks diluted in HCl	5	Activated Sludge	Anaerobic T = 35 °C pH 5.5	–	–	2.3	–	[70]
Microalgal biomass	10	*Enterobacter aerogenes* ATCC 13048 Sputum	Aerobic T = 30 °C pH 7.0	–	–	20.9	–	[71]
Aspen wood chopsticks	1.23	Enriched culture from hot spring	Anaerobic T = 50 °C pH 7.0	–	–	195	–	[72]
Depackaged food wastes	25	Sewage sludge	Anaerobic T = 35 °C, pH 7.0	–	–	0.0014	–	[48]

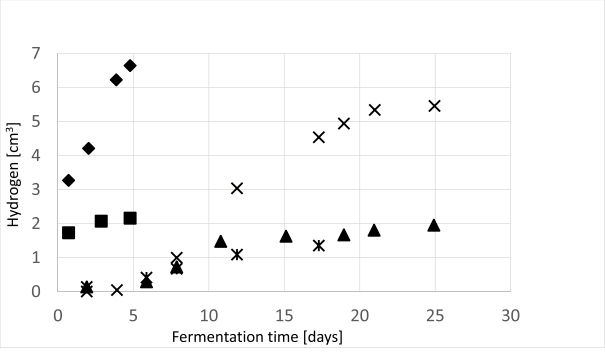


Fig. 5. Cumulative hydrogen production from cotton wastes and sour cabbage, under a load of 5 g VSS/L: ▲ cotton wastes OFR 1 mL/h, pH 6.23; ⋆ cotton wastes OFR 0.55 mL/h, pH 7.8; ◆ sour cabbage OFR 0.63 mL/h, pH 7.54; × crosses cotton wastes OFR 4.6 mL/h, pH 6.4; ■ sour cabbage anaerobic pH 7.54.

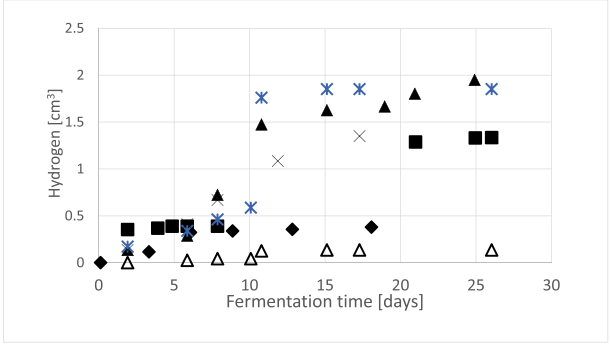


Fig. 6. Cumulative hydrogen production from cotton wastes, under a load of 5 g VSS/L: ◆ anaerobic pH 7.8; ⋆ OFR 0.55 mL/h, pH 7.8; ▲OFR 2.7 mL/h, pH 6.2; ■ OFR 1 mL/h pH, 6.23; × OFR 1.5 mL/h pH 7.84, Δ OFR 2.1 mL/h, pH 7.68.

day does not occur before the methane peak day as is usually the case [73,74]. From Table 2, it may be observed that food wastes such as potato waste or cabbage waste are better hydrogen sources with lower pretreatment demands – only milling – while cotton waste needs to be milled and hydrolysed. With the same pretreatments, wheat straw produces much less hydrogen – for high hydrogen production from it, improved lignin separation and hydrolysis is required. Therefore, efficient gas production is more cost effective using food wastes, rather than cotton waste, and the most expensive substrates are lignocellulose wastes which have a high percentage of lignin such as wheat straw [75]. We may observe that from 1 g of cotton under anaerobic conditions, more methane can be produced than from diary manure [62], food waste [64], switch grass [65] or sour cabbage [45].

Cotton waste sourced from old clothes is a very poor source of hydrogen when compared to the production rates from other substrates like barley straw [76], sugar beets [77], cotton stalk [10] or rice straw [78]. In agreement with earlier data [46] microaeration leads to a decrease in methane production and an increase in fermentation time, but it also increases hydrogen production. In contrast, microaeration

does not influence the fermentation time of sour cabbage.'

The hydrogen yield from cotton waste fermentation without pretreatment is much lower than that obtained for cotton stalk (containing more lignin that the cotton waste) - see Li et al. [10] but those latter results were obtained for a very specific inoculum (derived from carp intestine); a yield equal to 71% of the theoretical hydrogen potential was reported.

In light of the findings of ref. [2] and the results presented in Table 2, it may be stated that the hydrogen production yield for untreated inoculum is a viable parameter for choosing the substrate for commercial hydrogen production before stressing the inoculum.

Fig. 7 presents hydrogen yields versus oxygen flow rates. The existence of an optimal OFR for hydrogen production is clearly presented. For the substrate load considered here (5 g VSS/L) and neutral pH value (~7.5) the optimal value of OFR was around 0,6 mL/h both for cotton waste and sour cabbage [45], which corresponds to a hydrogen yield of 0,38 and 1,32 mL/g VSS for cotton waste and sour cabbage, respectively. When comparing hydrogen production for different OFRs, it may be observed (see Figs. 5–6) that except for a quite unusual one for 0.55 mL/h, the hydrogen yield for cotton waste is lower than that for sour cabbage by a factor of between 5 and 25. The existence of a range of OFR in which hydrogen production is higher than under anaerobic conditions may be observed. The OFR range suitable for hydrogen production is much wider for sour cabbage than for cotton waste. Hydrogen production from cotton waste occurs under neutral pH conditions for an OFR of between 0 and 2.6 mL/h, while for sour cabbage the range of OFR enabling hydrogen production extents up to 4 mL/h.

Decreasing the pH values extended the OFR range enabling hydrogen production, at least for cotton waste. Moreover, the optimal OFR value and hydrogen yield significantly increases with decrease of pH value, e.g. for pH 6.3, OFR = 4.6 mL/h and 1.1 mL/g VSS.

In contrast, methane production steadily decreases with OFR increase – see Fig. 8. This proves that in the case of dark fermentation, competitive reactions related to methanogenesis and hydrogenesis are taking place. The range of OFR which enable methane production is again wider in the case of sour cabbage than cotton waste. For OFR from 0 to 1.7 mL/h, methane production from cotton waste is higher than from sour cabbage. The maximum rate of methane production is achieved for anaerobic conditions; as OFR increases, production steadily decreases but not so rapidly as for hydrogen production. However, for sour cabbage the dependence in the OLR range from 0 to 3 mL/h is not very significant.

In relation to the mechanism of observed phenomenon it is possible that a small amount of oxygen can activate the immune systems of the bacteria [79], which inhibits methane production but allows for higher hydrogen production, as in the case of white rot stressed with silver nanoparticles [58,80]. Oxidative stress first inhibits methane production, which consumes the hydrogen produced; this liberates hydrogen otherwise consumed for methane generation [81]. Later, with growing stress exposure hydrogen overproduction is inhibited [79], which favours methane production processes [58,82,83].

As DF of cotton waste does not lead to high levels of hydrogen production, contrary to the prevailing opinions, an appropriate substrate most probably should also contain other lignocellulose material besides cellulose and hemicellulose [35].

4. Conclusions

Cotton waste is a potential source of methane, but is substantially less efficient than potato waste. Under anaerobic conditions and a load of 5 g VSS/L, methane production amounted to 0,78 L/g VSS over 17.1 days (and a methane content of up to 73.8%). The highest yield of hydrogen 1.1 mL/g VSS was obtained under acidic pH value of 6.4 and OFR (4.6 mL/h).

Oxygen plays the role of a stress factor for methanogenic bacteria, and when carefully controlled, it may lead to stable hydrogen production via dark fermentation. Moreover, it was found that there is an optimal value of OFR. The OFR ~0.6 mL/h is optimal for hydrogen production from cotton waste under neutral pH conditions (the same for sour cabbage). The optimal value of OFR increases for more acidic conditions (4.6 mL/h for pH value of 6.4). As increased microaeration leads to decreased methane production, the process may be used to optimize hydrogen production. The mechanism was partly explained by stress of anaerobic (methanogenic and hydrogenic) bacteria.

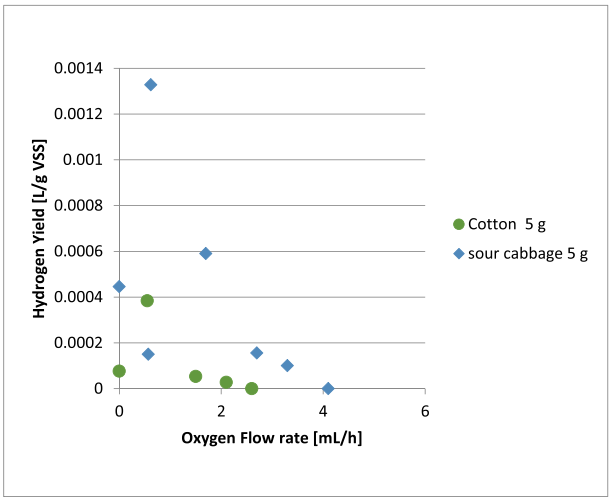


Fig. 7. Hydrogen yields in L/g VSS versus OFR under neutral pH conditions and a VSS load of 5 g/L for cotton wastes and sour cabbage.

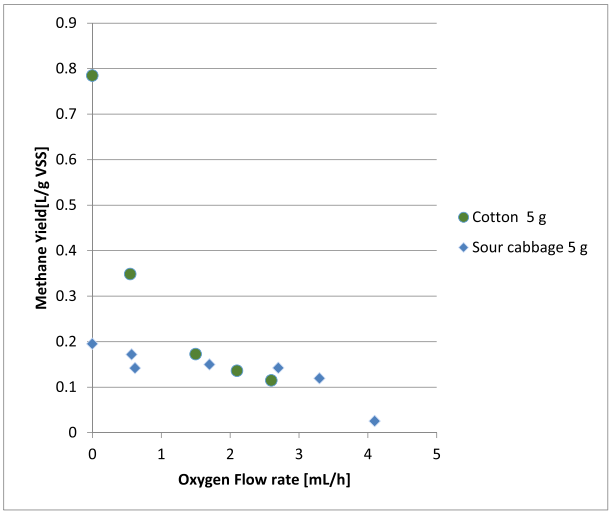


Fig. 8. Methane production versus OFR for a load of 5 g VSS/L from cotton wastes and sour cabbage.

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